IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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- Methodology
- Results
- Conclusion
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Executive Summary

Summary of methodologies

-Data Collection through API

-Data Collection with Web Scraping

-Data Wrangling

-Exploratory Data Analysis with SQL

-Exploratory Data Analysis with Data Visualization

-Interactive Visual Analytics with Folium

-Machine Learning Prediction

•Summary of all results

-Exploratory Data Analysis result

-Interactive analytics in screenshots

-Predictive Analytics result

Introduction

Project background and context

Space X advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because Space X can reuse the first stage. Therefore, if we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against space X for a rocket launch. This goal of the project is to create a machine learning pipeline to predict if the first stage will land successfully.

Problems you want to find answers

-What factors determine if the rocket will land successfully?

-The interaction amongst various features that determine the success rate of a successful landing.

-What operating conditions needs to be in place to ensure a successful landing program.

Section 1

Methodology

Methodology

Executive Summary

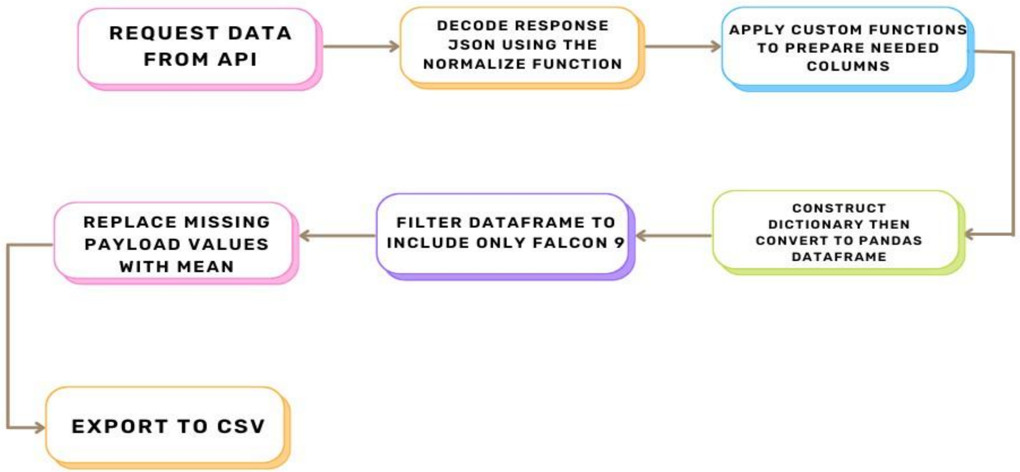
- Data collection methodology:
  - Data was collected using SpaceX API and web scraping from Wikipedia.
- Perform data wrangling
  - One-hot encoding was applied to categorical features
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

Data Collection

- The data was collected using various methods
  - Data collection was done using get request to the SpaceX API.
  - Next, we decoded the response content as a Json using .json() function call and turn it into a pandas dataframe using .json_normalize().
  - We then cleaned the data, checked for missing values and fill in missing values where necessary.
  - In addition, we performed web scraping from Wikipedia for Falcon 9 launch records with BeautifulSoup.
  - The objective was to extract the launch records as HTML table, parse the table and convert it to a pandas dataframe for future analysis.

Data Collection – SpaceX API

- We used the get request to the SpaceX API to collect data, clean the requested data and did some basic data wrangling and formatting.



1. Get request for rocket launch data using API

In [6]:
```
spacex_url="https://api.spacexdata.com/v4/launches/past"
```

In [7]:
```
response = requests.get(spacex_url)
```

2. Use json_normalize method to convert json result to dataframe

In [12]:
```
# Use json_normalize method to convert the json result into a dataframe

# decode response content as json
static_json_df = res.json()
```

In [13]:
```
# apply json_normalize
data = pd.json_normalize(static_json_df)
```

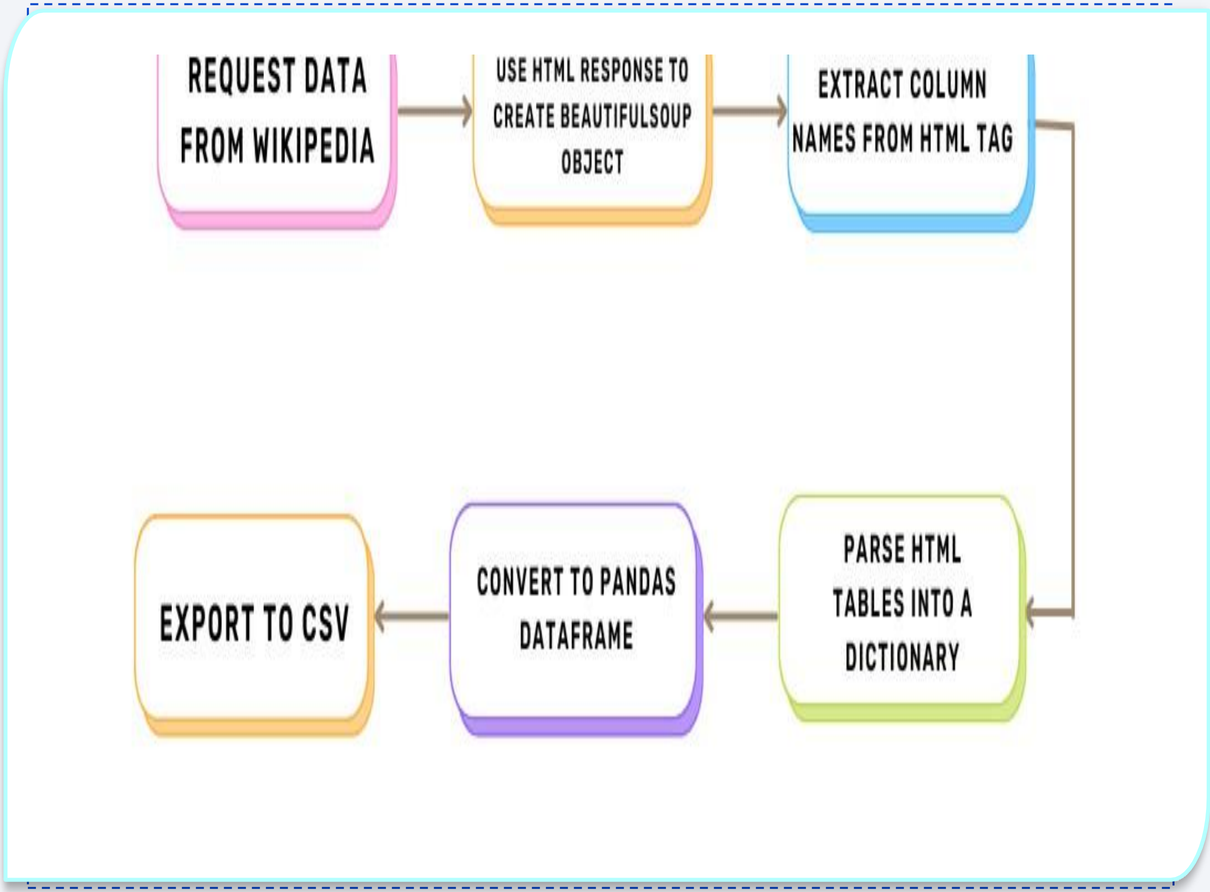
3. We then performed data cleaning and filling in the missing values

In [30]:
```
rows = data_falcon9['PayloadMass'].values.tolist()[0]

df_rows = pd.DataFrame(rows)
df_rows = df_rows.replace(np.nan, PayloadMass)

data_falcon9['PayloadMass'][0] = df_rows.values
data_falcon9
```

https://github.com/denji77/IBM-Applied-Data-Science-Capstone-project/blob/main/1.Data%20Collection%20API.ipynb

Data Collection - Scraping

- I applied web scrapping to webscrap Falcon 9 launch records with BeautifulSoup
- I parsed the table and converted it into a pandas dataframe.
- Then export to csv.



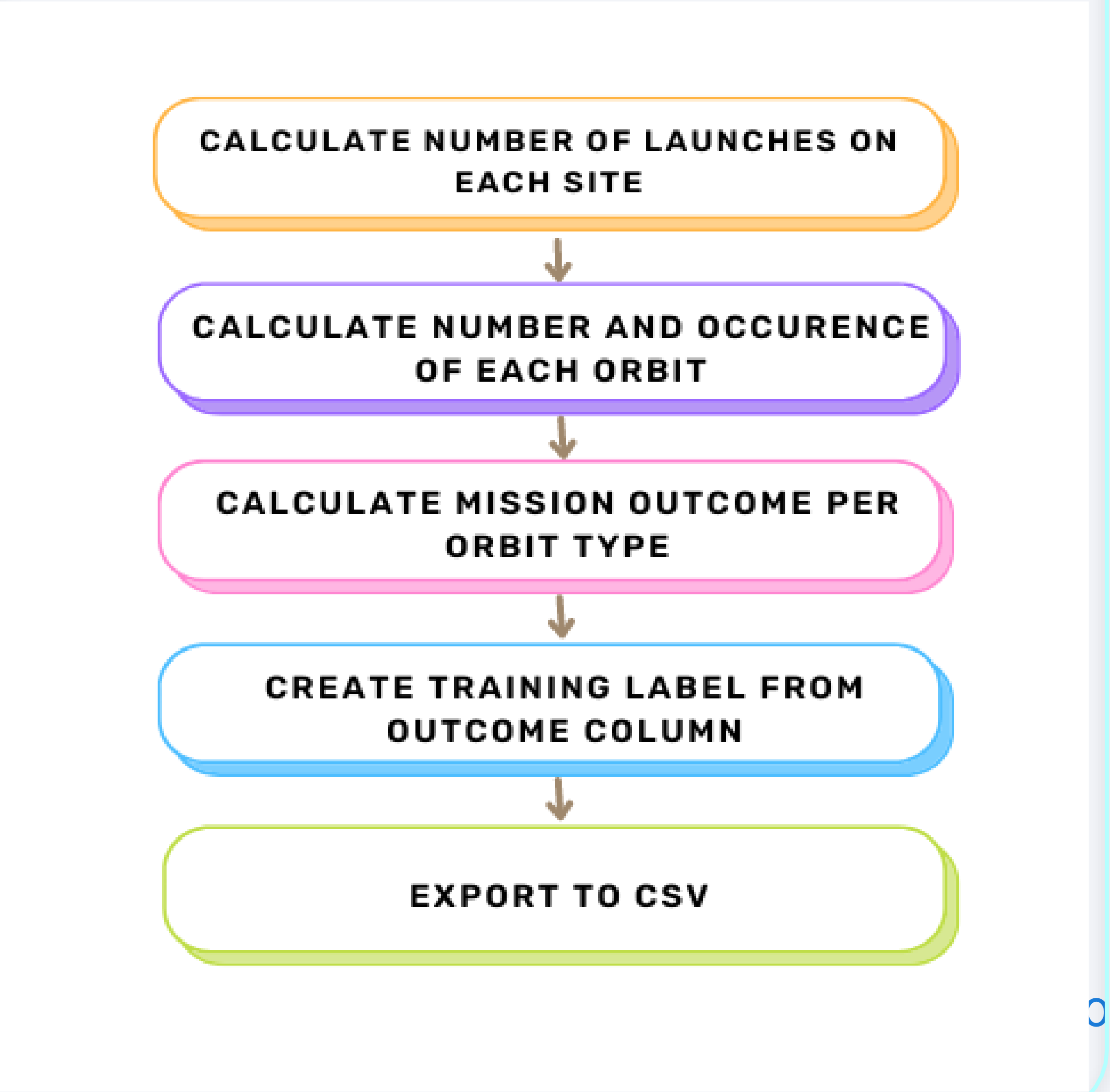
https://github.com/denji77/IBM-Applied-Data-Science-Capstone-project/blob/main/2.Data%20Collection%20with%20Web%20Scraping.ipynb

Data Wrangling

Booster Landing Outcomes: The dataset encompasses various booster landing scenarios: - Successful landings: "True Ocean," "True RTLS," "True ASDS" - Unsuccessful landings: "False Ocean," "False RTLS," "False ASDS"

For model training, these outcomes are transformed into binary labels: "1" signifies successful landing, "0" signifies unsuccessfullanding.

https://github.com/denji77/IBM-Applied-Data-Science-Capstone-project/blob/main/3.EDA.ipynb



EDA with Data Visualization

The analysis utilized various chart types to explore relationships within the data.

- Scatter Plots: Flight Number vs. Payload Mass, Flight Number vs. Launch Site, Payload Mass vs. Launch Site, Flight Number vs. Orbit Type, Payload Mass vs Orbit Type. These visualizations aim to identify potential correlations between the listed variables, which could be leveraged for machine learning model development.
- Bar Charts: Orbit Type vs. Success Rate. This chart focuses on comparing success rates across different orbital categories.
- Line Charts: Success Rate Yearly Trend. This visualization depicts the trend in launch success rate over time, allowing for an assessment of potential temporal patterns.

https://github.com/denji77/IBM-Applied-Data-Science-Capstone-project/blob/main/5.EDA%20Visualization.ipynb

EDA with SQL

- Unique Launch Sites: Identified the distinct launch site names used throughout the space missions.
- Launch Sites Starting with 'CCA': Retrieved the details of the first 5 launch missions conducted from sites whose names begin with the string "CCA."
- Payload Mass by Customer (NASA CRS): Calculated the total payload mass carried by boosters launched for NASA's CRS program.
- Average Payload Mass by Booster Version (F9 v1.1): Determined the average payload mass carried by the F9 v1.1 booster version.
- First Successful Ground Pad Landing: Identified the date of the first successful mission where the booster landed on a ground pad.
- Successful Drone Ship Landings (4000kg < Payload < 6000kg): Retrieved the names of boosters that successfully landed on a drone ship while carrying a payload mass between 4,000 kg and 6,000 kg.
- Mission Outcome Counts: Summarized the total number of successful and failed missions.
- Maximum Payload Mass by Booster Version: Identified the booster versions responsible for carrying the heaviest payloads.
- Failed Drone Ship Landings (2015): Listed the failed landing outcomes in a drone ship for the year 2015, including the corresponding booster versions and launch site names.
- Landing Outcome Ranking (2010-06-04 to 2017-03-20): Ranked the frequency of various landing outcomes (e.g., Failure (drone ship), Success (ground pad)) in descending order for launches occurring between June 4th, 2010, and March 20th, 2017.

- https://github.com/denji77/IBM-Applied-Data-Science-Capstone-project/blob/main/4.EDA%20with%20SQL.ipynb

Build an Interactive Map with Folium

- We marked all launch sites, and added map objects such as markers, circles, lines to mark the success or failure of launches for each site on the folium map.
- We assigned the feature launch outcomes (failure or success) to class 0 and 1.i.e., 0 for failure, and 1 for success.
- Using the color-labeled marker clusters, we identified which launch sites have relatively high success rate.
- We calculated the distances between a launch site to its proximities. We answered some question for instance:
  - Are launch sites near railways, highways and coastlines.
  - Do launch sites keep certain distance away from cities.

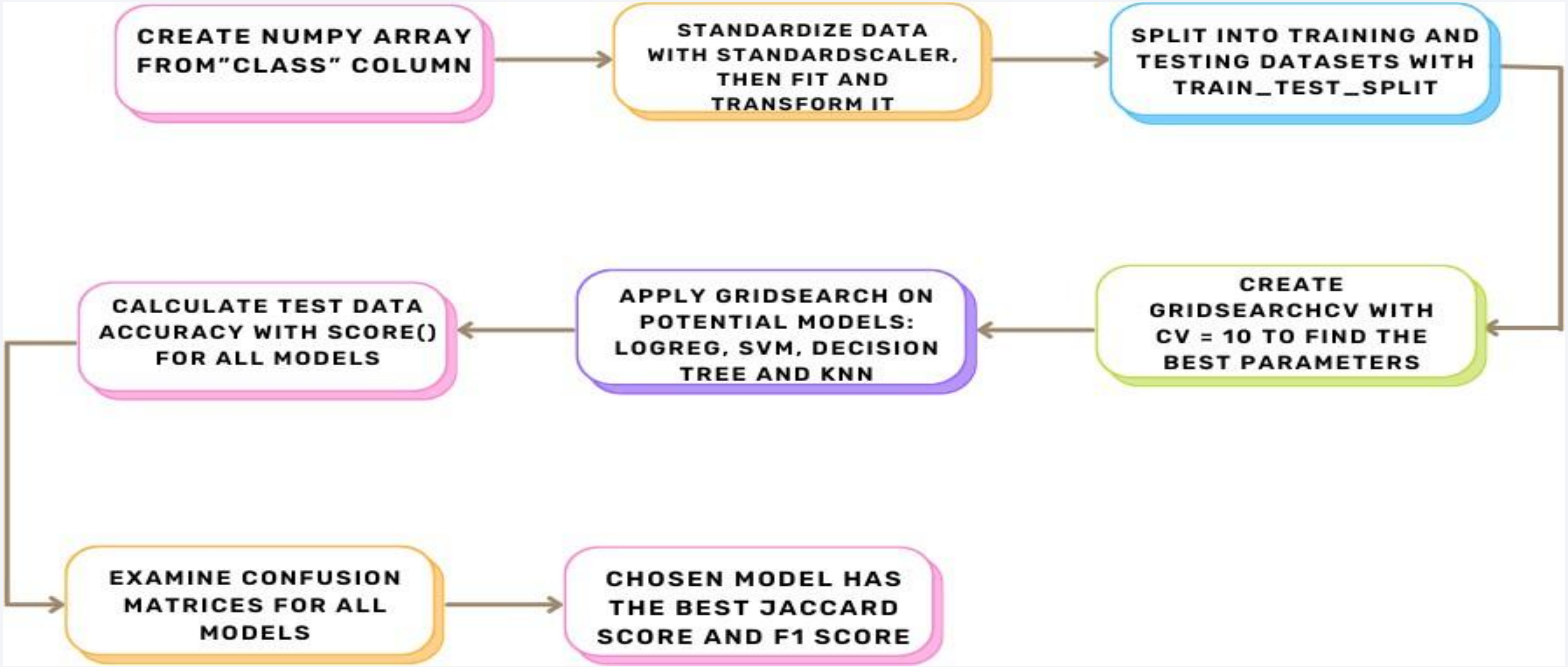
- GITHUB LINK
- https://github.com/denji77/IBM-Applied-Data-Science-Capstone-project/blob/main/6.Interactive%20Visual%20Analytics%20with%20Folium%20lab.ipynb

Build a Dashboard with Plotly Dash

- **Interactive Launch Site Selection:** A dropdown list was implemented to facilitate the selection of a specific launch site for analysis.
- **Dynamic Pie Chart:** A pie chart displays launch success data. When "All Sites" is selected, the chart presents the total successful launches across all sites. Upon selecting a specific launch site, the chart dynamically updates to show the success and failure breakdown for that chosen site.
- **Payload Mass Range Selection:** A slider allows users to define a desired payload mass range for further investigation.
- **Payload Mass vs. Success Rate Scatter Chart:** A scatter chart visually depicts the correlation between payload mass and launch success rate for the various booster versions employed. This visualization helps identify potential relationships between payload weight and launch outcomes.

https://github.com/denji77/IBM-Applied-Data-Science-Capstone-project/blob/main/spacex%20dash%20app.py

Predictive Analysis (Classification)



Results

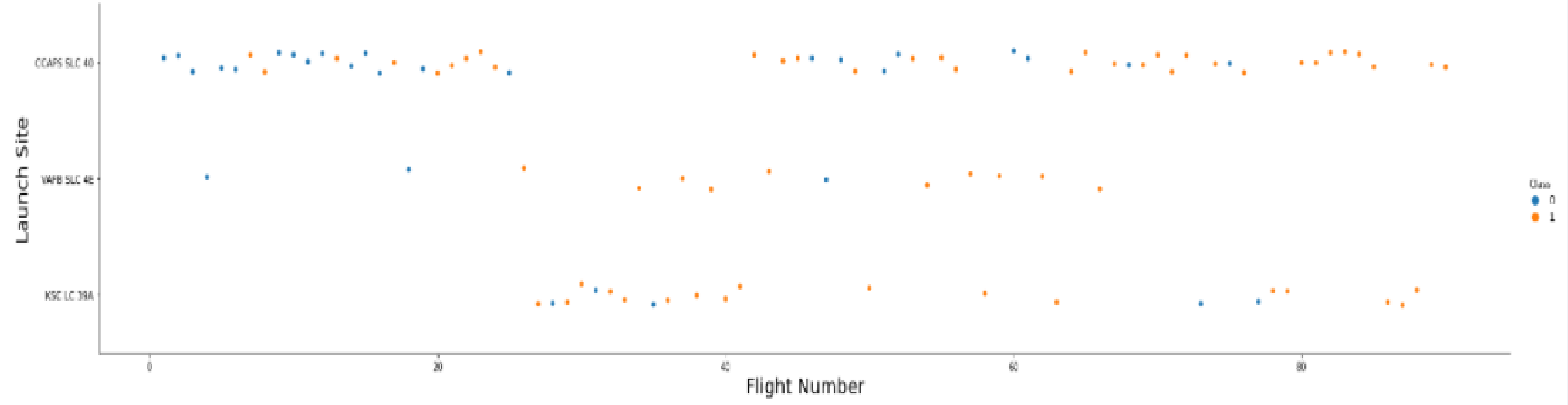
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

Insights drawn from EDA

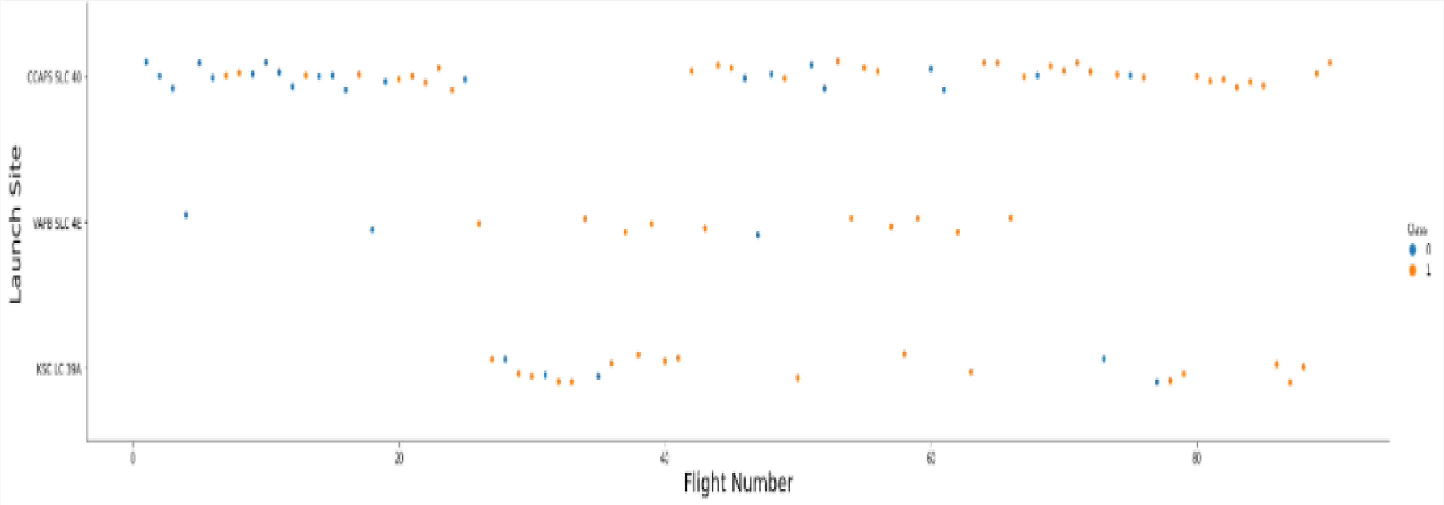
Flight Number vs. Launch Site

- From the plot, we found that the larger the flight amount at a launch site, the greater the success rate at a launch site.



Payload vs. Launch Site

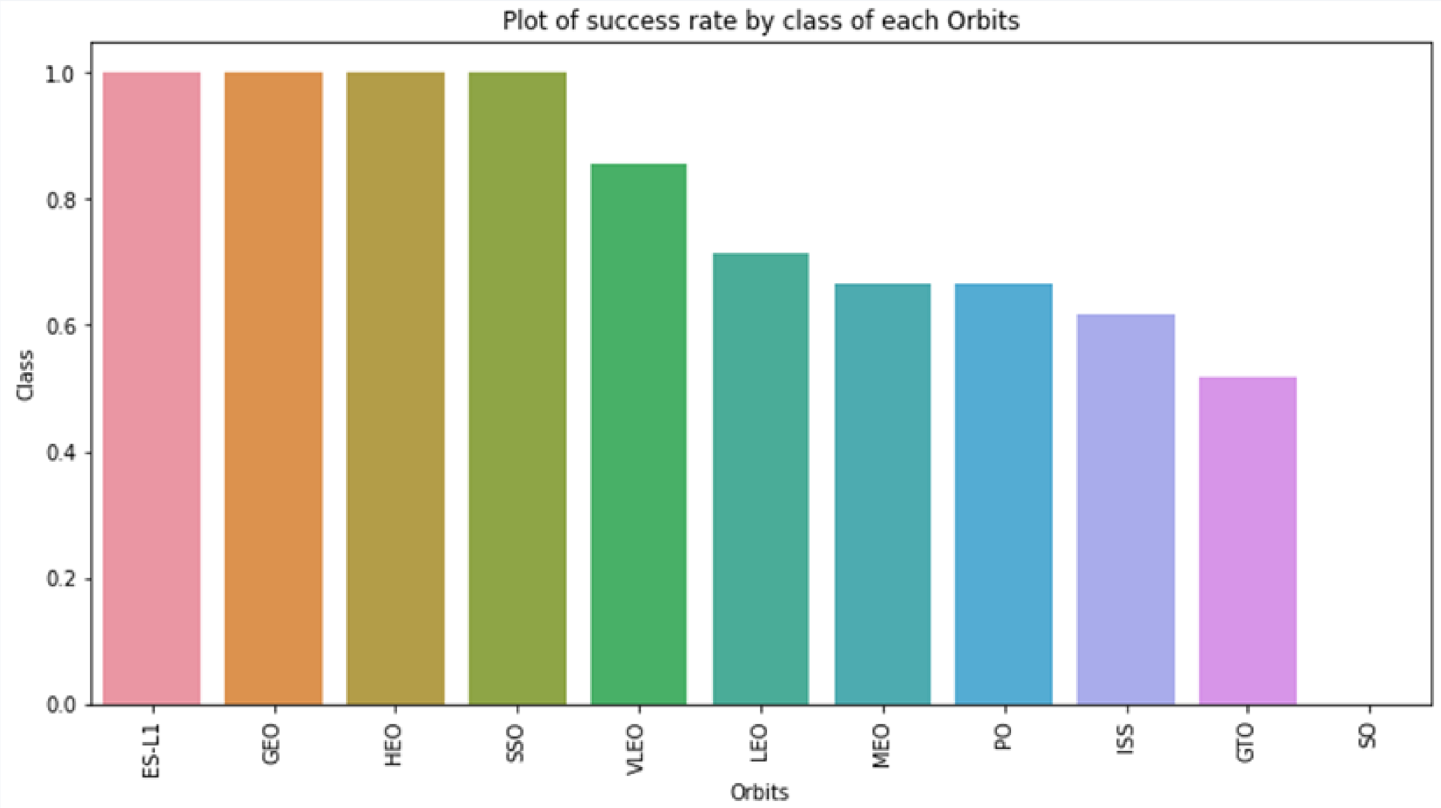
- The analysis suggests a potential positive correlation between payload mass and launch success rate across various launch sites. Launches exceeding 7,000 kg in payload mass appear to have a higher success rate overall.
- Notably, KSC LC-39A exhibits a 100% success rate for missions carrying payloads under 5,500 kg.



Success Rate vs. Orbit Type

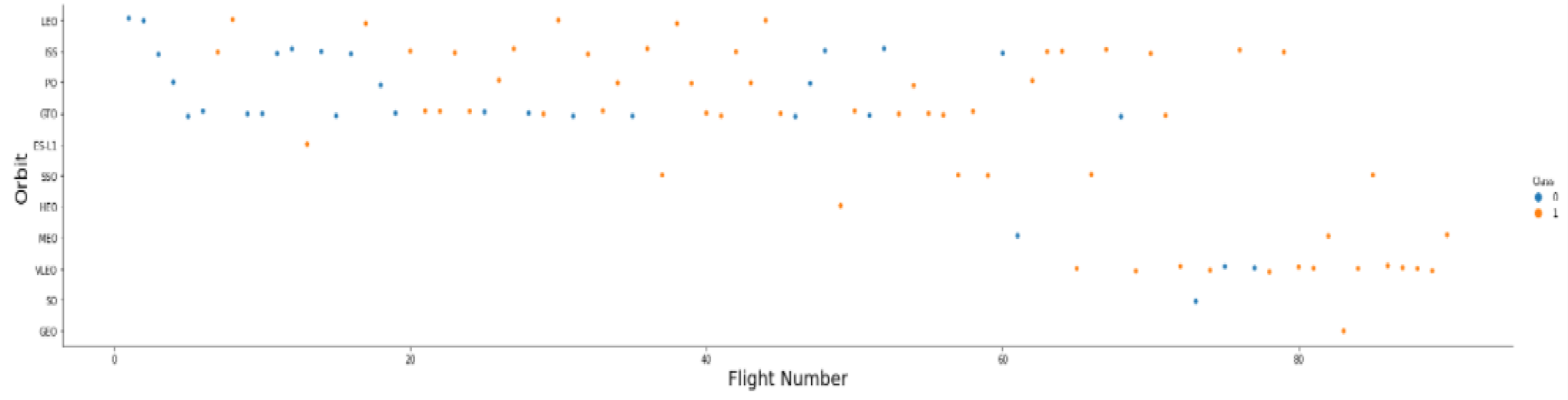
The analysis revealed varying success rates across different orbital categories:

- **100% Success Rate:** Orbits designated ES-L1, GEO, HEO, and SSO exhibited a perfect success record within the dataset.
- **0% Success Rate:** Launches targeting SO orbit did not achieve success in any instances.
- **Moderate Success Rates (50% - 85%):** GTO, ISS, LEO, MEO, and PO orbits demonstrated success rates ranging from 50% to 85%.



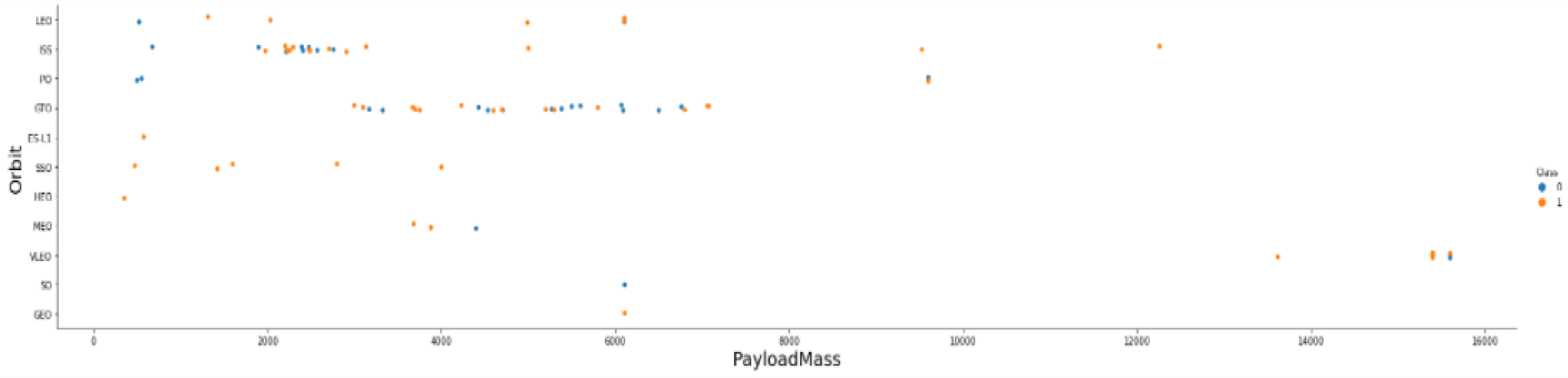
Flight Number vs. Orbit Type

- The plot below shows the Flight Number vs. Orbit type. We observe that in the LEO orbit, success is related to the number of flights whereas in the GTO orbit, there is no relationship between flight number and the orbit.



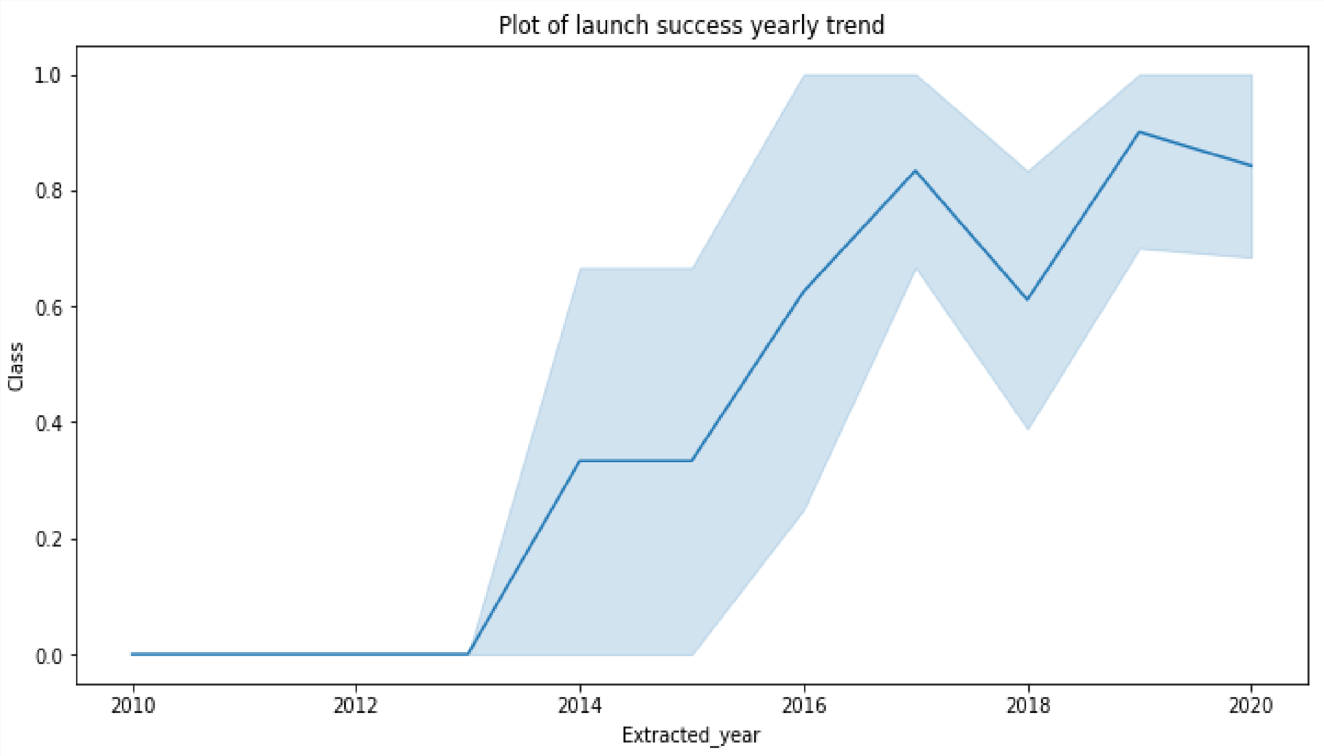
Payload vs. Orbit Type

- We can observe that with heavy payloads, the successful landing are more for PO, LEO and ISS orbits.



Launch Success Yearly Trend

- From the plot, we can observe that success rate since 2013 kept on increasing till 2020.
- The success rate rose steadily but not linearly from 2013.



All Launch Site Names

- Displays the names of the unique launch sites.
- We can see that there are four (4).

Display the names of the unique launch sites in the space mission

In [10]:

```
task_1 = '''
        SELECT DISTINCT LaunchSite
        FROM SpaceX
'''
create_pandas_df(task_1, database=conn)
```

Out[10]:

	launchsite
0	KSC LC-39A
1	CCAFS LC-40
2	CCAFS SLC-40
3	VAFB SLC-4E

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`

Display 5 records where launch sites begin with the string 'CCA'

```
%sql select * from SPACEXTABLE where launch_site like 'CCA%' limit 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Calculate the total payload carried by boosters from NASA

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql select sum(payload_mass__kg_) as totalMass from SPACEXTABLE where customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

totalMass
45596

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- We calculated the average payload mass carried by booster version F9 v1.1 as 2928.4.

Display average payload mass carried by booster version F9 v1.1

In [13]:

```
task_4 = '''
        SELECT AVG(PayloadMassKG) AS Avg_PayloadMass
        FROM SpaceX
        WHERE BoosterVersion = 'F9 v1.1'
        '''
create_pandas_df(task_4, database=conn)
```

Out[13]:

	avg_payloadmass
0	2928.4

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad.
- We observed that the dates of the first successful landing outcome on ground pad was 22nd December 2015

In [14]:
```
task_5 = '''
        SELECT MIN(Date) AS FirstSuccessfull_landing_date
        FROM SpaceX
        WHERE LandingOutcome LIKE 'Success (ground pad)'
        '''
create_pandas_df(task_5, database=conn)
```

Out[14]:

	firstsuccessfull_landing_date
0	2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

We used the **WHERE** clause to filter for boosters which have successfully landed on drone ship and applied the **AND** condition to determine successful landing with payload mass greater than 4000 but less than 6000

In [15]:

```
task_6 = '''
        SELECT BoosterVersion
        FROM SpaceX
        WHERE LandingOutcome = 'Success (drone ship)'
            AND PayloadMassKG > 4000
            AND PayloadMassKG < 6000
        '''
create_pandas_df(task_6, database=conn)
```

Out[15]:

	boosterversion
0	F9 FT B1022
1	F9 FT B1026
2	F9 FT B1021.2
3	F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
%sql select mission_outcome, count(*) as total_number from SPACEXTABLE group by mission_outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	total_number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- We determined the booster that have carried the maximum payload using a subquery in the **WHERE** clause and the **MAX()** function.

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

In [17]:

```
task_8 = '''
        SELECT BoosterVersion, PayloadMassKG
        FROM SpaceX
        WHERE PayloadMassKG = (
                                SELECT MAX(PayloadMassKG)
                                FROM SpaceX
                                )
        ORDER BY BoosterVersion
        '''
create_pandas_df(task_8, database=conn)
```

Out[17]:

	boosterversion	payloadmasskg
0	F9 B5 B1048.4	15600
1	F9 B5 B1048.5	15600
2	F9 B5 B1049.4	15600
3	F9 B5 B1049.5	15600
4	F9 B5 B1049.7	15600
5	F9 B5 B1051.3	15600
6	F9 B5 B1051.4	15600
7	F9 B5 B1051.6	15600
8	F9 B5 B1056.4	15600
9	F9 B5 B1058.3	15600
10	F9 B5 B1060.2	15600
11	F9 B5 B1060.3	15600

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
%%sql select substr(Date, 6, 2) as month, date, booster_version, launch_site, landing_outcome from SPACEXTABLE
    where landing_outcome = 'Failure (drone ship)' and substr(Date, 0, 5) = '2015';
```

```
 * sqlite:///my_data1.db
Done.
```

month	Date	Booster_Version	Launch_Site	Landing_Outcome
01	2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%%sql select landing_outcome, count(*) as count_outcomes from SPACEXTABLE
      where date between '2010-06-04' and '2017-03-20'
      group by landing_outcome
      order by count_outcomes desc;
```

```
 * sqlite:///my_data1.db
Done.
```

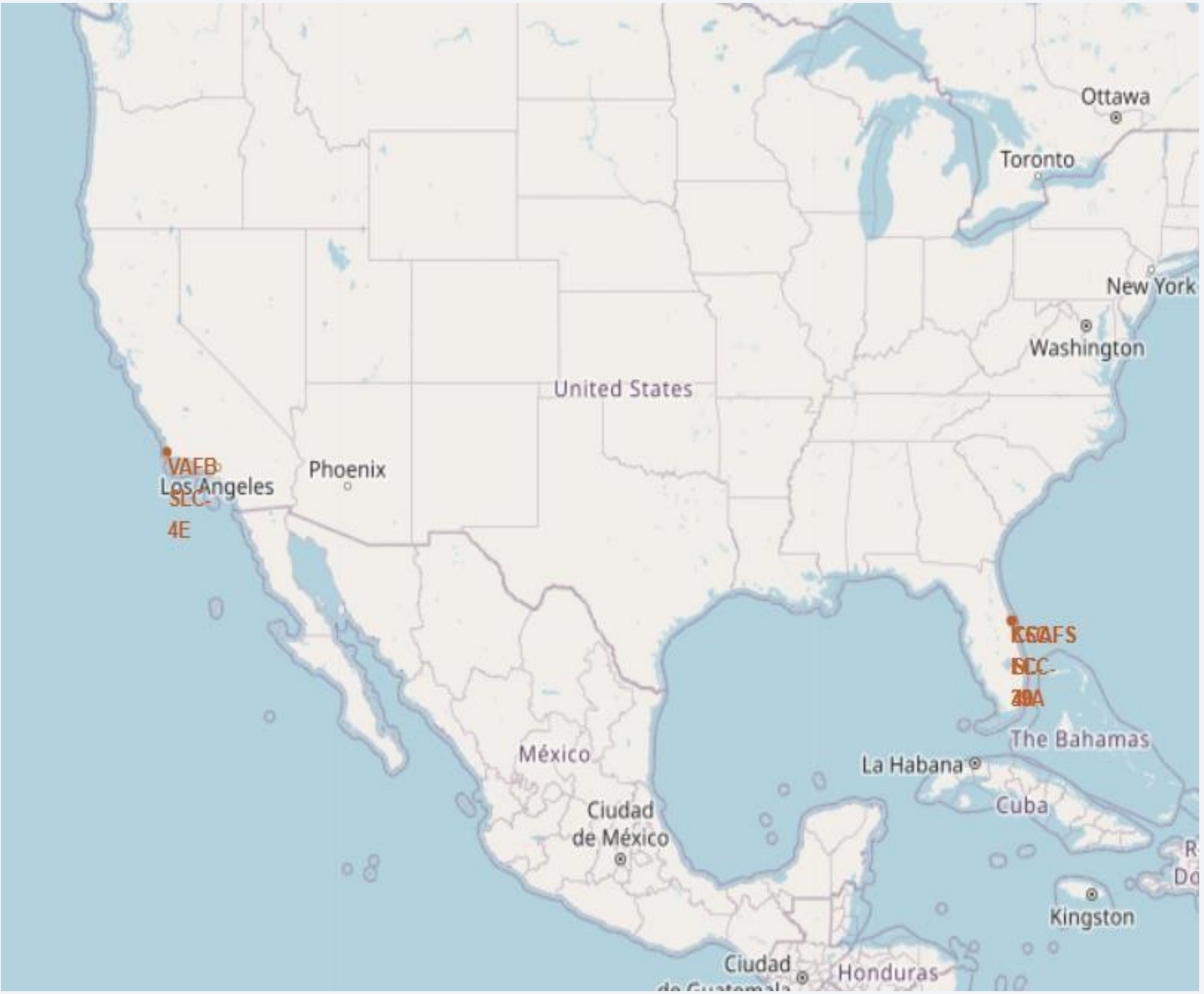
Landing_Outcome	count_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

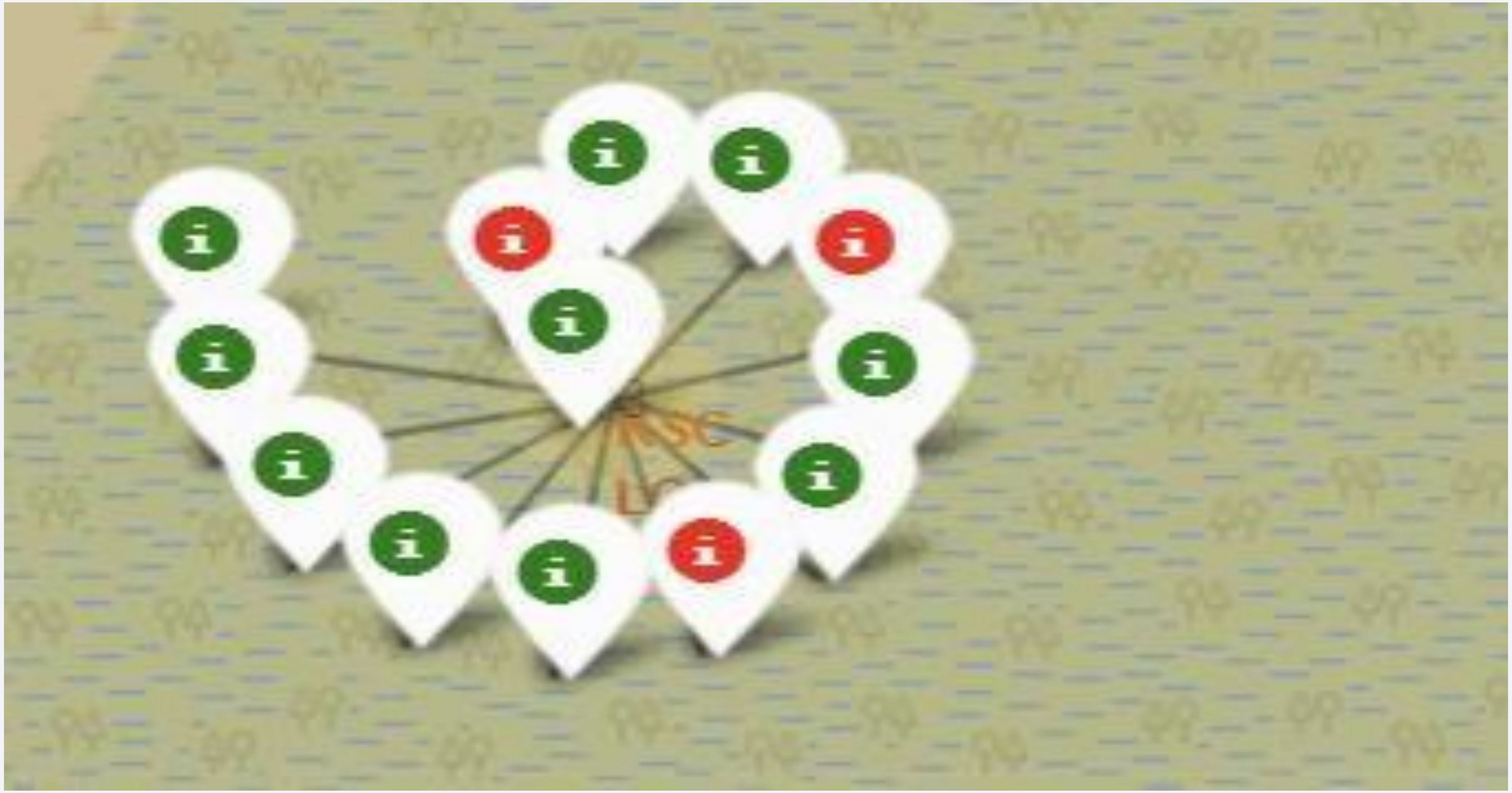
Launch Sites Proximities Analysis

All launch sites global map markers

- The analysis reveals a concentration of launch sites near the Earth's equator. At the equator, the Earth's surface has the highest linear velocity. A spacecraft launched from the equator inherits this eastward velocity, providing an initial boost in orbital speed and reducing the fuel required to achieve orbit.
- The clustering of launch sites close to coastlines offers a safety advantage. In the event of a launch mishap, debris dispersion is directed away from populated areas and towards the ocean, minimizing potential risks on land.



Markers showing launch sites with color labels

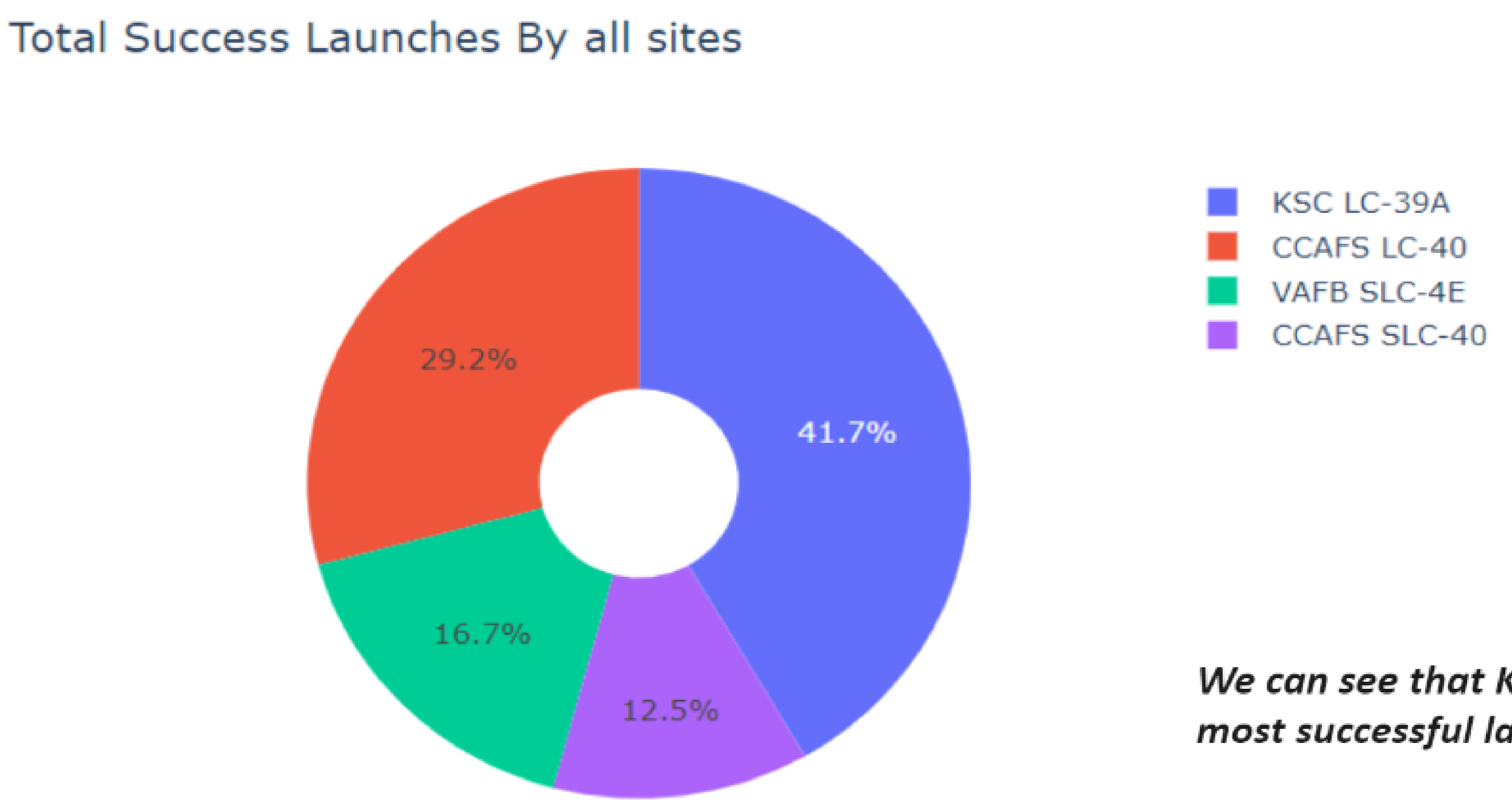


Launch Site distance to landmarks

Section 4

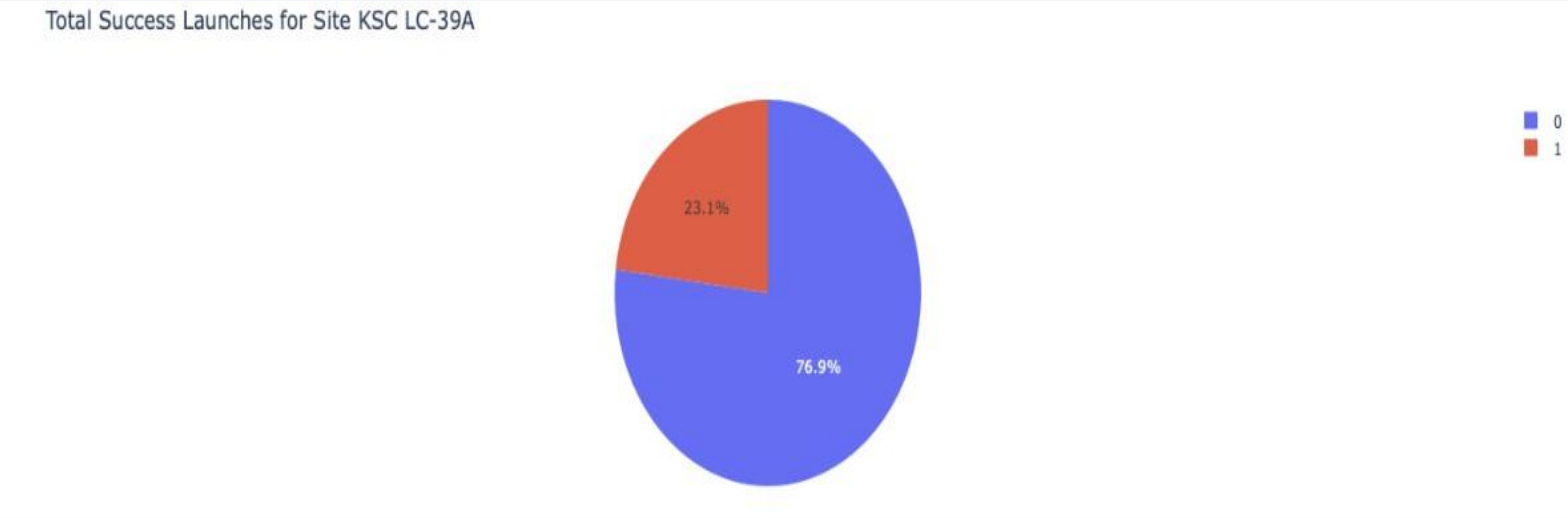
Build a Dashboard with Plotly Dash

Success percentage achieved by each launch site

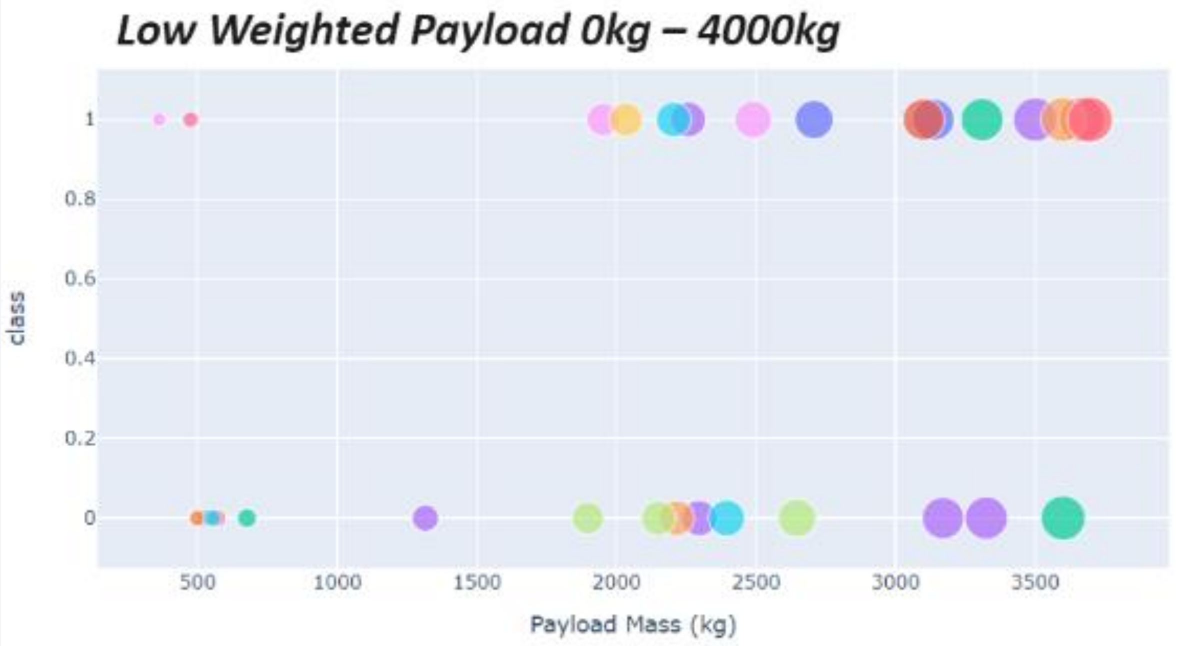


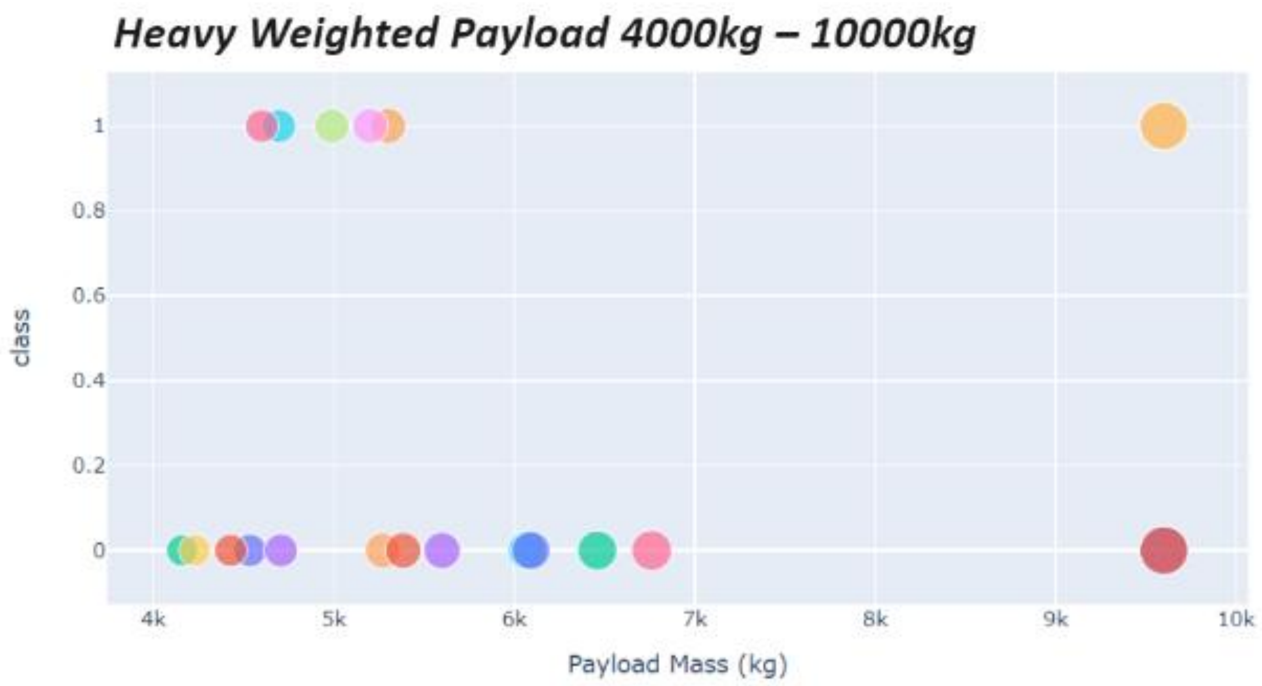
We can see that KSC LC-39A had the most successful launches from all the sites

Launch site with the highest launch success ratio



Scatter plot of Payload vs Launch Outcome for all sites, with different payload selected in the range slider





We can see the success rates for low weighted payloads is higher than the heavy weighted payloads

Section 5

Predictive Analysis (Classification)

Classification Accuracy

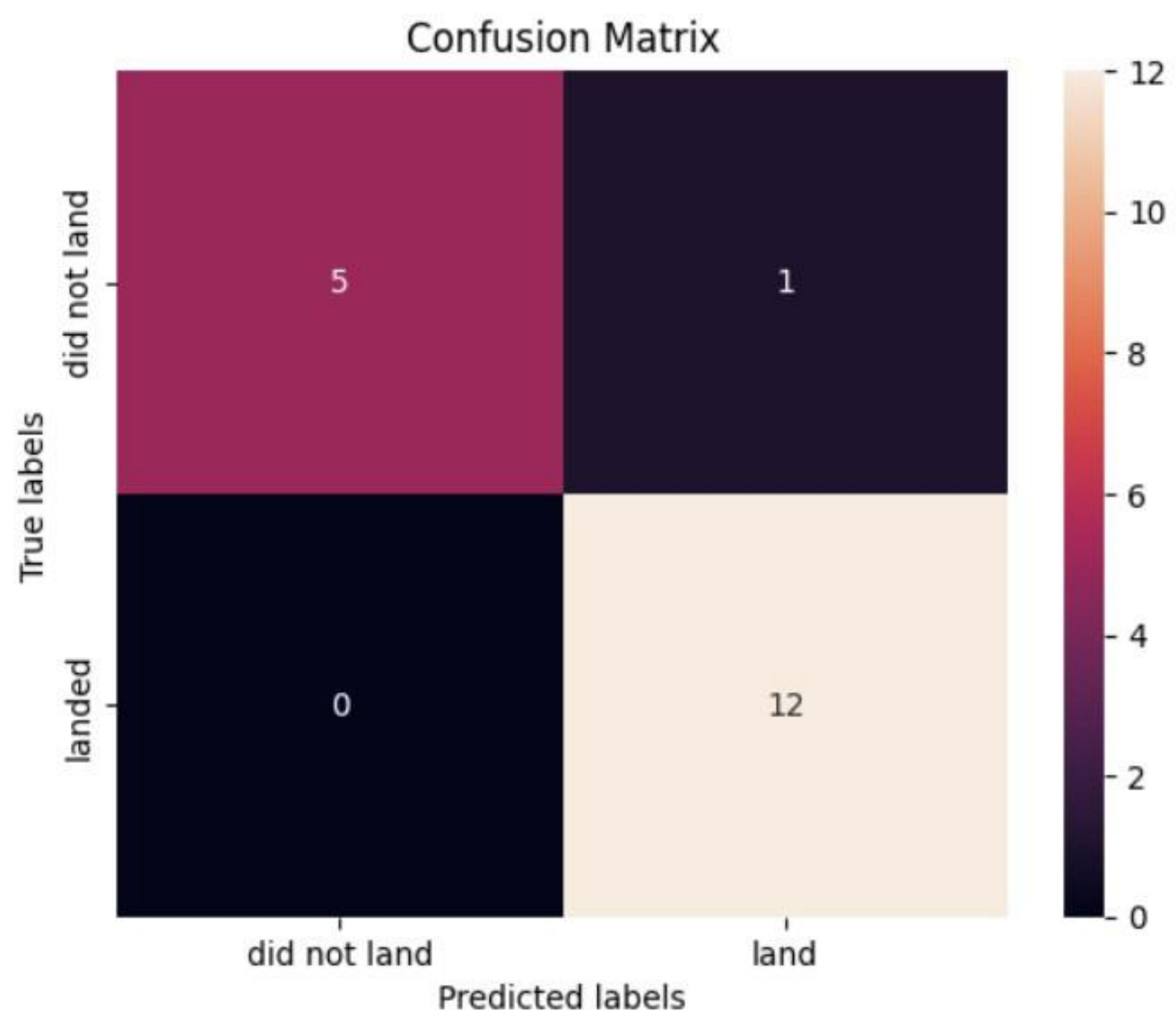
The decision tree classifier is the model with the highest classification accuracy

```
models = {'KNeighbors':knn_cv.best_score_,
              'DecisionTree':tree_cv.best_score_,
              'LogisticRegression':logreg_cv.best_score_,
              'SupportVector': svm_cv.best_score_}

bestalgorithm = max(models, key=models.get)
print('Best model is', bestalgorithm,'with a score of', models[bestalgorithm])
if bestalgorithm == 'DecisionTree':
    print('Best params is :', tree_cv.best_params_)
if bestalgorithm == 'KNeighbors':
    print('Best params is :', knn_cv.best_params_)
if bestalgorithm == 'LogisticRegression':
    print('Best params is :', logreg_cv.best_params_)
if bestalgorithm == 'SupportVector':
    print('Best params is :', svm_cv.best_params_)
```

```
Best model is DecisionTree with a score of 0.8732142857142856
Best params is : {'criterion': 'gini', 'max_depth': 6, 'max_features': 'auto', 'min_samples_leaf': 2, 'min_samples_split': 5, 'splitter': 'random'}
```

Confusion Matrix

- The confusion matrix for the decision tree classifier shows that the classifier can distinguish between the different classes. The major problem is the false positives .i.e., unsuccessful landing marked as successful landing by the classifier.



Conclusions

- The analysis determined that the decision tree algorithm is the most appropriate model for this dataset based on its performance metrics.
- A noticeable pattern emerged, indicating that launches with lighter payloads generally achieve higher success rates compared to those carrying heavier payloads.
- Launch sites tend to cluster strategically near the equator, taking advantage of the Earth's rotational speed to enhance launch efficiency. Their closeness to coastlines also helps minimize risks in the event of launch failures.
- The study indicated a gradual increase in launch success rates over time, reflecting advancements in technology and improvements in launch procedures.
- Among all launch sites, KSC LC-39A stood out with the highest overall success rate in the dataset. Additionally, certain orbits—namely ES-L1, GEO, HEO, and SSO—demonstrated a flawless success record based on the analyzed data.

Thank you!